

LAB 3

Database Integration & Server-Side Programming

Lab Title

Database Design and Backend Development using PHP & MySQL

1. Lab Overview

In this lab, students will implement the **backend functionality** required for their semester project.

Students will create:

- database tables
- PHP database connection
- item management functionality
- server-side form processing

2. Learning Objectives

Students will learn to:

- connect PHP to MySQL
- design relational database tables
- implement CRUD operations
- store and retrieve data from the database

3. Prerequisites

Students must complete **Lab 2**.

Students must understand:

- basic PHP syntax
- SQL commands

4. Lab Tasks

Task 1 — Create Database

Create database: `project_database`

Task 2 — Create Core Tables

Create the following tables:

Users

`users`

Items

`items`

Categories

`categories`

Example SQL:

```
CREATE TABLE categories (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100)  
);  
  
CREATE TABLE items (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(150),  
  price DECIMAL(10,2),  
  category_id INT,  
  description TEXT  
);
```

Task 3 — Create Database Connection

Create file: `config/database.php`

Example:

```
$conn = new mysqli("localhost","root","","project_database");
```

Task 4 — Create Item Submission Form

Create page: `create_item.php`

Form fields:

- item name
 - price
 - category
 - description
-

Task 5 — Insert Data into Database

Create file: `save_item.php`

Insert data using SQL:

```
INSERT INTO items (name,price,category_id,description)
VALUES (?, ?, ?, ?)
```

Task 6 — Display Items from Database

Create page: `items.php`

Retrieve data:

```
SELECT * FROM items
```

Display results as item cards.

5. Expected Results

Students should have:

- working database
- items stored in database
- dynamic item catalog page
- PHP database connection working

6. Deliverables

Students demonstrate:

- database tables
- item submission form
- item catalog page